
Semantic-Aware Logit Adjustment for Long-Tailed Recognition

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Abstract

Training deep neural networks on imbalanced data is difficult as conventional training methods exhibit bias towards the head classes. Existing methods address long-tailed bias issue through frequency correction or representation learning, but they do not quantify class-level feature overlap and optimize the loss accordingly. In this work, we introduce Semantic-Aware Logit Adjustment (SALA), a training loss which integrates the concept of frequency priors together with semantic crowdedness penalty based on the similarities of the classifiers' prototypes. Our method not only considers the rarity of each class, but also the degree of overlap of the representations between the classes to optimize the most structurally crowded classes. Extensive experiments performed on standard benchmarks like ImageNet-LT, CIFAR-10-LT, CIFAR-100-LT and iNaturalist 2018 show highly competitive performance, particularly excelling in rare-class accuracy.

1 Introduction

Deep learning models have been very successful in dealing with well-curated datasets with even class distributions [36]. However, in practice, such as in applications like self-driving cars [27], medical diagnoses [17], and social media analysis [40] often exhibit an inherent class imbalance, where a handful of head classes comprise the majority of the data, while many tail classes lack representation. Deep learning models trained using regular cross-entropy often experience severe performance drops for the tail classes.

To address this performance deterioration, the initial set of methods concentrate on resampling the dataset [12, 3, 11, 9] and cost-sensitive re-weighting [26, 7, 32, 46]. While resampling attempts to equalize class distributions through over-sampling of the tail or under-sampling of the head class,

it often results in a substantial risk of overfitting to the minority class or wasteful utilization of the abundant majority class samples. In contrast, re-weighting approaches increase penalties for tail class errors, which could lead to optimization instabilities and cause degradation in accuracy for the majority classes. The critical problem with this type of approach is that it tries to influence the training process without solving the bias issue that arises from the imbalance itself. More recent methods focus on modifying the class margins [6, 19, 8, 29] or reshaping class geometry [39, 52, 4, 23, 47], while some rely on contrastive objectives [25, 53, 10, 15] to improve inter-class separation. However, these approaches largely treat class imbalance and representation separation as broadly shared across the classes without explicitly accounting for the varying degree of feature overlap between the classes. In practice, semantically similar rare classes require greater attention so this motivates a more adaptive mechanism that accounts for both class rarity and semantic crowdedness.

To address these concerns, we introduce Semantic-Aware Logit Adjustment which accounts for both class rarity and semantic crowdedness. This design provides our framework two perspectives. First, it preserves the benefits of frequency-aware correction while avoiding the limitation of treating all classes as equally difficult. Second, it introduces an explicit mechanism for addressing the residual semantic bias that persists even after frequency adjustment. By penalizing the crowded classes more strongly, the proposed approach encourages the feature extractor to learn more discriminative representations for semantically similar classes. As a result, the learned features become more semantically organized, enabling minority classes gain clearer separation from visually similar neighbors while maintaining stable optimization by grounding the semantic penalty in the geometry of the current classifier rather than in a separate auxiliary objective.

The major contributions are:

- To propose Semantic-Aware Logit Adjustment (SALA), a new loss that incorporates both the class rarity and semantic crowdedness. By considering both aspects of the problem, we are able to prevent the excessive correction of semantically distinct classes while simultaneously optimizing the structurally crowded classes.
- SALA is a lightweight, plug-and-play loss-level modification, seamlessly integrated into existing long-tailed recognition pipelines without requiring additional networks, contrastive branches, or complex training architectures.
- To perform experiments on benchmark datasets, such as ImageNet-LT [35], iNaturalist 2018 [14], CIFAR-10-LT, CIFAR-100-LT [21]. Our proposed method demonstrates its effectiveness by achieving competitive overall performance and substantial improvements specifically for rare classes.

2 Related Works

2.1 Re-sampling and Re-weighting Methods

Traditional approaches for fixing class imbalance either modify how the training data is distributed or adjust loss function weights. To mitigate class imbalance, re-sampling techniques can be employed to over-sample minority classes or under-sample majority classes [3, 30, 12, 11, 9]. However, over-sampling minority class can lead to overfitting, while under-sampling majority class results in the loss of potentially important information. Alternatively, re-weighting assigns greater importance to the minority classes by adjusting their weight during loss calculation [26, 7, 32, 46]. This method preserves the cardinality of the dataset but it can destabilize optimization, particularly under severe class imbalance where gradients from rare classes may become disproportionately dominant.

2.2 Logit Adjustment and Margin-Based Approaches

Margin-based methods [6, 19, 8] tweak the decision boundary rather than the training data itself. For example, Equalization Loss [38], Seesaw Loss [41], and LDAM [2] use class-specific margins to create broader decision boundaries, especially for minority classes. Logit Adjustment (LA) [29] and Balanced Softmax [34] then offer a solid Bayesian backing for this approach, adjusting the logits by $\log \pi_c$ to directly address the shifts in class priors. More recently, researchers enhanced the LA framework with extra awareness terms. Take Adaptive Logit Adjustment (ALA) [50], which adjusts the frequency term based on instance difficulty. Then there’s Difficulty-aware Balancing Margin (DBM)

Loss [37], which modifies the margin penalty to handle different learning difficulties across classes. Likewise, Margin Calibration (MARC) [42] works on logits through affine transformations. Many recent works [39, 52, 4, 23, 47] see difficulty as variations or errors in statistics at the instance level. In contrast, SALA is different since it goes beyond these by assessing the geometric entanglement.

2.3 Contrastive Learning for Long-Tailed Recognition

Supervised Contrastive Learning (SupCon) [20] improves representation learning by pulling samples from the same class together while separating other classes. Under long-tailed distributions, head classes dominate contrastive interactions and bias the learned representations. Few works introduce the class- or structure-aware contrastive objectives, including Balanced Contrastive Learning (BCL) [53], Targeted Supervised Contrastive Learning (TSC) [25], and Subclass-balancing Contrastive Learning (SBCL) [15]. More recent methods further address this bias through decoupled objectives [33], probabilistic modeling of class distributions [10]. Despite aforementioned advances, existing methods mainly modify sample-level contrastive interactions or their optimization. But SALA explicitly measures the semantic crowdedness of class prototypes and incorporates this class-specific geometry into logit adjustment which is just a plug-and-play modification.

3 Methodology

3.1 Preliminary

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ be a long-tailed dataset where $y_i \in [C] := \{1, \dots, C\}$. The sample counts n_c follow $n_1 \geq \dots \geq n_C > 0$ and the empirical class prior is $\pi_c = n_c/N$. A network maps inputs to features $h = f_\theta(x) \in \mathbb{R}^d$ and logits $z = Wh \in \mathbb{R}^C$, where W contains prototypes $w_c^\top$. We denote L_2 -normalized vectors as $\bar{w}_c = w_c/\|w_c\|_2$.

The standard cross-entropy (CE) loss is expressed by:

$$\mathcal{L}_{\text{CE}}(z, y) = -\log \frac{\exp(z_y)}{\sum_{j=1}^C \exp(z_j)} \quad (1)$$

Standard CE optimization works great with balanced datasets, but it tends to model the skewed training prior $\mathbb{P}(y) \approx \pi_y$. This becomes a concern in long-tailed distributions where it favors head classes, leaving tail classes behind. Menon et al. [29] came up with Logit Adjustment (LA) technique to fix this issue. Inspired by statistical decision theory, LA tweaks the logits based on class priors to even out the error rates and give greater importance to the rare classes. Their method, which focuses on increasing the margin for less common classes, is summarized by the LA loss formula:

$$\mathcal{L}_{\text{LA}}(z, y) = -\log \frac{\exp(z_y + \tau \log \pi_y)}{\sum_{j=1}^C \exp(z_j + \tau \log \pi_j)} \quad (2)$$

where $\tau > 0$ is a scaling parameter (often set to $\tau = 1$).

To understand the exact mechanics of this adjustment on head versus tail classes, by considering gradient of $\mathcal{L}_{\text{LA}}$ w.r.t. the target logit z_y . Let p_y^{LA} show the adjusted softmax probability for the target class derived from the Equation 3.1. The gradient is given by:

$$\frac{\partial \mathcal{L}_{\text{LA}}}{\partial z_y} = p_y^{\text{LA}} - 1 \quad (3)$$

Let us consider two training samples, one from head class and another from tail class, they both yield the exact same raw target logit z . Due to the frequency-based priors,

$$z + \tau \log \pi_{y_t} < z + \tau \log \pi_{y_h} \quad (4)$$

Consequently, the adjusted probability for the tail class is strictly suppressed compared to head class ($p_{y_t}^{\text{LA}} < p_{y_h}^{\text{LA}}$). Because the magnitude of the gradient updating the logit is $|p_y^{\text{LA}} - 1| = 1 - p_y^{\text{LA}}$, the tail sample produces a larger gradient scale:

$$\left| \frac{\partial \mathcal{L}_{\text{LA}}}{\partial z_{y_t}} \right| > \left| \frac{\partial \mathcal{L}_{\text{LA}}}{\partial z_{y_h}} \right| \quad (5)$$

This enables the model to adjust its focus dynamically, prioritizing the optimization for harder, underrepresented classes. It pushes for bigger gradient steps where they are needed the most.

We analyze the geometry of the learned classifier through the Neural Collapse (NC) framework [31]. In terminal phase of training, NC1 suggests that within-class features become tightly concentrated around their class means, while NC3 indicates an alignment between the classifier weights and the corresponding class means. Accordingly, for a sample (x, y) , its feature representation can be approximated as

$$h(x) \approx s \cdot \bar{w}_y, \quad (6)$$

where $s > 0$ denotes the feature norm and $\bar{w}_y$ is the normalized prototype of the target class. Under this approximation, the logit corresponding to class c becomes

$$z_c(x) = w_c^\top h(x) \approx s \|w_c\|_2 (\bar{w}_c^\top \bar{w}_y). \quad (7)$$

Thus, the relative logits are directly determined by the angular relationships between the class prototypes. In an ideally separated representation, non-target prototypes have low cosine similarity with the target prototype. But under long-tailed training, this geometry can become distorted, mainly for the minority classes whose prototypes may become closely aligned with the dominant classes. Such positive prototype overlap increases the competition faced by the corresponding class and leaves a residual geometric bias that frequency-based adjustment alone cannot capture.

3.2 Semantic-Aware Logit Adjustment (SALA)

We introduce Semantic-Aware Logit Adjustment (SALA) which penalizes classes based on feature overlap in latent space rather than training data rarity alone. This allows the model to determine how difficult it is to distinguish between classes during learning independently of how frequently each class appear.

The Semantic Crowdedness Score Ω_c measures how tightly packed and intertwined a specific class is on the unit hypersphere. By the Neural Collapse approximation of Eq. (6), the normalized prototypes $\bar{w}_c = w_c / \|w_c\|_2$ from Section 3.1 represent the class-mean directions in feature space so crowdedness measured on prototypes is crowdedness measured on the underlying class clusters. Thus, we quantify the alignment between any two classes c and k with the same cosine similarity term that drove the logit competition in Eq. (7), $\bar{w}_c^\top \bar{w}_k = \cos(\theta_{c,k})$, where $\theta_{c,k}$ is the angle between the two prototypes. We define the semantic crowdedness score as:

$$\Omega_c = \sum_{k \neq c} \max(0, \bar{w}_c^\top \bar{w}_k) \quad (8)$$

This formulation introduces two main mathematical design rules. First, the rectified linear unit $\max(0, \cdot)$ retains only the active geometric interference. Classes separated by an angle of 90° or more do not affect each other since their similarity is zero. But for angles under 90° , the impact grows with how close the classes are. Second, Ω_c deliberately excludes the empirical frequency prior $\pi_k = n_k/N$ that drives $\mathcal{L}_{LA}$ in Eq. (3.1). It serves as a pure measure of how similar the classes are, orthogonal to the frequency term it sits alongside in Eq. (9) below. This makes it extra sensitive to those rare class entanglements where sample counts almost match up ($\pi_{y_t}/\pi_k \rightarrow 1$).

We integrate this semantic term directly into the logit adjustment framework of Eq. (3.1) to penalize the crowded classes during optimization. The final SALA loss simply augments $\mathcal{L}_{LA}$ with the $-\alpha\Omega_y$ term:

$$\mathcal{L}_{SALA}(z, y) = -\log \frac{\exp(z_y + \tau \log \pi_y - \alpha\Omega_y)}{\sum_{j=1}^C \exp(z_j + \tau \log \pi_j - \alpha\Omega_j)} \quad (9)$$

Here $\tau > 0$ retains its role from Eq. (3.1), controlling the frequency prior adjustment, while $\alpha > 0$ is a new scaling hyperparameter for the semantic penalty. Subtracting $\alpha\Omega_y$ from the target logit reduces the model’s confidence for crowded classes. These changes encourage the feature extractor to move overlapping class representations away from each other.

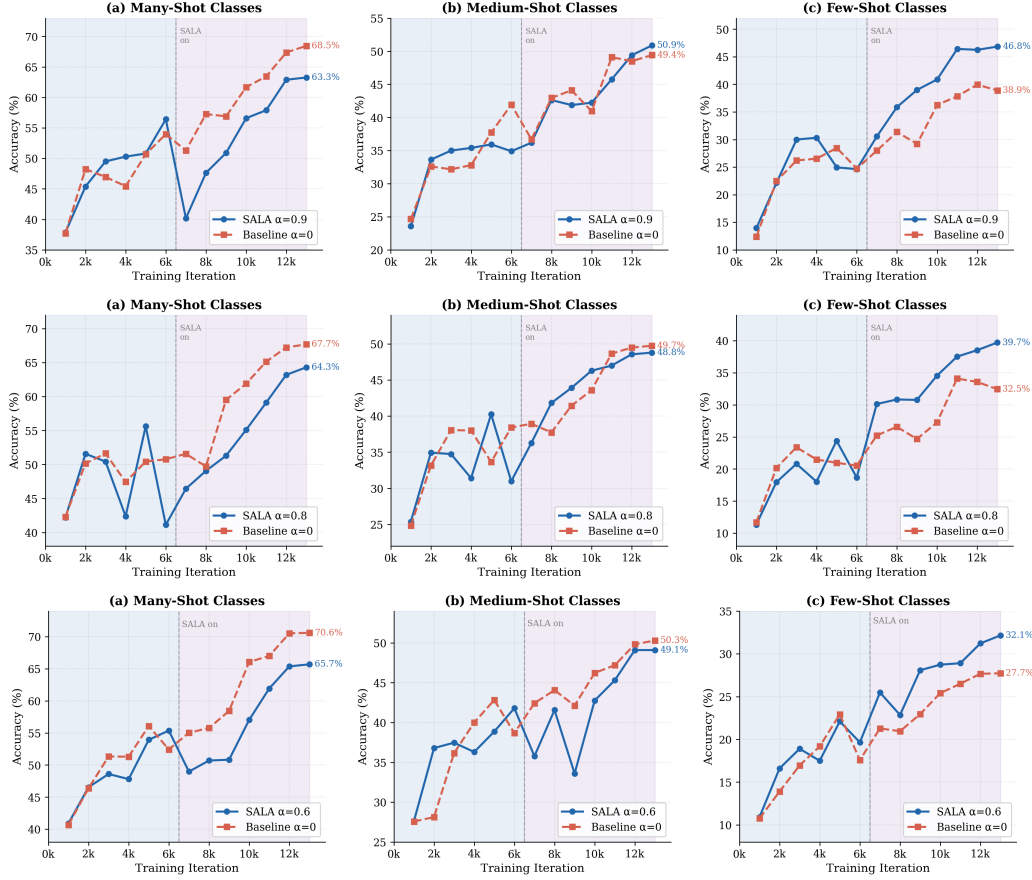


Figure 1: Split-wise testing accuracy over training iterations on CIFAR-100-LT for imbalance ratios 50, 100, and 200, respectively, where the blue line denotes the method with the semantic crowdedness term and the orange line denotes the variant with just the frequency term.

4 Experiments

4.1 Datasets

The proposed SALA loss is assessed on four commonly employed benchmarks for the task of long-tailed visual recognition, which include CIFAR-10-LT [21], CIFAR-100-LT [21], ImageNet-LT [28], and iNaturalist2018 [14]. The benchmark datasets involve long-tailed synthetic and real-world data at both small-scale and large-scale levels, providing an extensive experiment under different levels of imbalance severity. For the CIFAR-10-LT and CIFAR-100-LT datasets, long-tailed versions of the datasets are obtained by exponential undersampling their counterparts. Three levels of imbalance factor (IF) are considered, that are $\{50, 100, 200\}$ where IF represents the ratio of the most common class to the least common class. On the other hand, ImageNet-LT and iNaturalist2018 are large-scale naturally imbalanced datasets exhibiting highly skewed class distributions. Dataset statistics are summarized in Table 2.

4.2 Implementation Details

The same training protocol from Ren et al. [34] is followed for all experiments. We use Nesterov SGD with a momentum of 0.9 and weight decay of 5×10^{-4} for all model weights. For CIFAR-10-LT and CIFAR-100-LT, we use ResNet-32 as backbone, with a mini-batch size of 512. The learning rate starts at 0.05, increasing linearly to 0.1 over the first 800 iterations and then follows a cosine annealing schedule for 200 epochs, eventually dropping to 0. For iNaturalist 2018 and ImageNet-LT, we switch

Table 1: Top-1 accuracy (%) on CIFAR-10-LT and CIFAR-100-LT using ResNet32.

Dataset	CIFAR-10-LT			CIFAR-100-LT		
Imbalance Factor	50	100	200	50	100	200
Cross-Entropy	–	77.40	71.20	43.90	38.43	34.84
Logit Adjustment [29]	83.98	81.03	77.97	52.11	46.90	43.36
LDAM-DRW [2]	81.03	77.03	73.52	47.62	42.04	38.91
Balanced Softmax [34]	–	82.90	75.50	50.50	45.00	43.30
cRT [18]	84.21	79.15	73.06	50.86	45.12	41.73
Weight Balancing [1]	–	–	–	49.80	44.30	40.80
MisLAS [51]	85.70	82.10	–	52.30	47.00	42.33
GCL [22]	85.46	82.68	79.03	53.55	48.71	44.88
MetaSAug [24]	84.03	80.54	76.89	51.90	46.87	39.94
SWRM [5]	–	–	–	52.01	46.80	–
LADE [13]	–	–	–	50.50	45.40	–
ResLT [16]	–	83.46	80.44	–	49.98	45.34
BKD [49]	85.10	82.50	–	51.70	46.50	–
TSC [25]	82.90	79.70	–	47.40	43.80	–
SETF [45]	83.80	79.70	74.20	52.50	48.30	43.00
ARBloss [43]	86.10	83.80	79.50	49.70	47.10	42.70
CAL [44]	85.90	84.10	80.00	54.30	50.10	46.50
SALA	86.25	84.96	81.11	55.19	51.41	47.30

Table 2: Datasets Statistics.

Dataset	Classes	Train	Test
CIFAR-10	10	50,000	10,000
CIFAR-100	100	50,000	10,000
ImageNet-LT	1,000	115,846	50,000
iNaturalist2018	8,142	437,513	24,426

to ResNet-50 and train from scratch for 90 epochs using the same optimizer setup. Experiments were conducted on the NVIDIA A100 GPU.

4.3 Main Results

CIFAR-10-LT. Regarding CIFAR-10-LT, SALA outperforms other existing approaches across all imbalance factors. When $IF = 50$, SALA achieves an accuracy of 86.25% by outperforming other robust alternatives like ARBloss (86.10%), MisLAS (85.70%), and GCL (85.46%). As the degree of imbalance increases, SALA consistently maintains the lead, by producing 84.96% and 81.11% accuracy for IF values of 100 and 200, respectively. These accuracies significantly outperform all the other competing techniques, where CAL reports accuracies of 84.10% and 80.00% for IF values of 100 and 200, respectively.

CIFAR-100-LT. SALA demonstrates even more significant improvements on the CIFAR-100-LT dataset. As we can see in Table 1 and from the split-wise learning curves in Fig. 1, SALA activation quickly leads to overall balanced improvements in performance. The proposed approach outperforms current approaches like CAL (54.30%), GCL (53.55%), MisLAS (52.30%), and SWRM (52.01%) for $IF = 50$. As imbalance factor increases, SALA continues to outperform its competitors. Specifically,

Table 3: Top-1 accuracy (%) on ImageNet-LT and iNat2018 using ResNet50.

Dataset	ImageNet-LT	iNat2018
Cross-Entropy	44.51	63.80
Logit Adjustment [29]	52.75	68.11
LDAM-DRW [2]	48.80	68.00
Balanced Softmax [34]	47.20	–
cRT [18]	50.30	65.20
MisLAS [51]	53.40	71.60
Seesaw Loss [41]	50.40	–
MetaSAug [24]	47.39	68.75
LADE [13]	53.00	70.00
GCL [22]	54.88	72.01
GALA [48]	53.40	71.20
ALA [50]	52.40	70.70
MARC [42]	52.30	70.40
SETF [45]	44.70	–
TSC [25]	52.40	69.70
ARBloss [43]	52.80	–
CAL [44]	49.70	–
SALA	56.02	74.34

SALA achieves accuracies of 51.41% at $IF = 100$ and 47.30% at $IF = 200$, both being competitive to other existing methods.

ImageNet-LT. We further investigate the performance of our method by conducting experiments on ImageNet-LT (Table 3) to examine the efficacy of our approach under the setting of extreme long-tailed class distributions. It is shown that SALA can achieve a Top-1 accuracy of 56.02%, which is significantly better than other strong baselines, that includes GCL (54.88%), MisLAS (53.40%), LADE (53.00%), and GALA (53.40%).

iNaturalist2018. We further evaluate our method on the iNaturalist2018 dataset. As we can see in Table 3 that our method obtains 74.34% Top-1 accuracy and outperforms other existing approaches like GCL (72.01%), MisLAS (71.60%) and GALA (71.20%). These improvements exhibit that our method can effectively address the overconfidence problem of tail classes while learning discriminative features for the fine-grained categories.

Table 4: Ablation study on CIFAR-100-LT (ResNet32). ✓ indicates that the term is active. Top-1 accuracy (%) reported.

Configuration	Freq. Term ($\tau \log \pi_c$)	Geom. Term ($\alpha \Omega_c$)	CIFAR-100-LT		
			IF=50	IF=100	IF=200
Baseline	×	×	43.90	38.43	34.84
Freq. Only	✓	×	52.11	46.90	43.36
Sem. Only	×	✓	49.73	44.85	41.57
SALA (Both)	✓	✓	55.19	51.41	47.30

4.4 Ablation Study

To isolate the contribution of each component in SALA, we evaluate four configurations on CIFAR-100-LT: (1) standard cross-entropy, (2) frequency prior only, (3) semantic crowdedness penalty only, and (4) SALA.

As shown in Table 4, no single component matches the full SALA objective. Freq. Only confirms the value of prior correction but leaves semantic crowdedness unresolved. Geom. Only improves over

the baseline, by demonstrating that Ω_c carries independent structural signal, yet lacks the frequency calibration needed for tail-vs-head separation. Their combination yields consistent additive gains that widen with imbalance severity by confirming that the two signals are complementary.

5 Conclusion

In this work, we propose Semantic-Aware Logit Adjustment (SALA) that augments the standard frequency-based correction with a semantic crowdedness penalty Ω_c computed directly from the cosine similarity between classifier prototypes. By jointly accounting for class rarity and semantic crowdedness, it selectively penalizes classes that are both underrepresented and structurally entangled with their neighbors rather than treating all the tail classes as uniformly difficult. The resulting objective is lightweight and plug-and-play, by requiring no auxiliary networks or contrastive branches.

Experiments on ImageNet-LT, CIFAR-10-LT, CIFAR-100-LT, and iNaturalist 2018 show that SALA consistently outperforms strong frequency-based, margin-based, and contrastive baselines, with especially strong gains on the more fine-grained datasets.

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A Computational Overhead Analysis

We analyze the computational cost of SALA relative to standard Cross-Entropy (CE) and Logit Adjustment (LA). SALA introduces a semantic penalty based on pairwise cosine similarities between classifier prototypes. Let C denote the number of classes, d the feature dimension, and B the mini-batch size.

Complexity. The standard CE loss requires a backbone forward pass followed by a linear classifier with a complexity $O(B \cdot C \cdot d)$. SALA adds two classifier-head operations: (i) L_2 -normalization of the classifier weights, which costs $O(C \cdot d)$, and (ii) construction of the prototype similarity matrix $S = \bar{W}\bar{W}^\top \in \mathbb{R}^{C \times C}$, which costs $O(C^2 \cdot d)$ when recomputed from scratch. In practice, these operations act only on the classifier head and are therefore much cheaper than the convolutional backbone.

Dataset-specific implementation. For CIFAR-10-LT, CIFAR-100-LT, and ImageNet-LT, the prototype similarity matrix is refreshed every iteration. For iNaturalist2018, where $C = 8,142$, we reduce the cost by **refreshing the prototype similarity matrix once every 100 iterations** and reusing the cached semantic scores between refreshes. This makes the expensive $O(C^2 \cdot d)$ computation amortized over multiple updates and prevents the semantic term from becoming a bottleneck on the largest benchmark.

Table 5: Computational overhead of SALA relative to the backbone forward pass. Overhead is reported per training iteration, with the iNaturalist2018 prototype similarity matrix refreshed every 100 iterations.

Dataset	Backbone	Classes (C)	Dim (d)	Refresh Rate	Backbone FLOPs	SALA Overhead
CIFAR-10-LT	ResNet-32	10	64	every iter.	≈ 0.07 G	$< 0.01\%$
CIFAR-100-LT	ResNet-32	100	64	every iter.	≈ 0.07 G	$\approx 0.08\%$
ImageNet-LT	ResNet-50	1,000	2,048	every iter.	≈ 3.80 G	$\approx 0.24\%$
iNaturalist18	ResNet-50	8,142	2,048	every 100 iters	≈ 3.80 G	$\approx 0.14\%$ (amortized)

Scalability. For extreme classification tasks where C may reach tens of thousands, recomputing the full similarity matrix at every step can become really expensive. SALA therefore incorporates periodic prototype updates, i.e., recomputing S once every K steps or once per epoch. This reduces the amortized similarity cost from $O(C^2 \cdot d)$ per iteration to $O(C^2 \cdot d/K)$, while preserving the geometric effect of the semantic bias.

Empirical observations. In our experiments on an NVIDIA A100 GPU, we observed no measurable difference in wall-clock training time per epoch between the baseline LA and SALA (± 1.2 seconds for ImageNet-LT). This indicates that SALA provides the intended geometric benefits without materially increasing the training budget.

B Training Procedure

We use two different optimization techniques to handle the semantic penalty and maintain the training stability. At first during the forward pass, we detach the classifier weights from the computational graph; this it fixes Ω_c as a constant scalar for that particular iteration. Making sure that $\nabla_W \Omega_c = 0$. Therefore, gradients only flow through the logit representations to update the network parameters and prevent chaotic optimization loops. Second, we follow a two-stage training process where initially the model runs with $\alpha = 0$, which lets class clusters form based on simple frequency counts. When these initial spaces become stable, we then turn on the SALA penalty to untangle the fine-grained geometric issues affecting the tail classes.